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RESEARCH ARTICLE

Edge ML Technique for Smart Traffic Management in Intelligent Transportation Systems

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ABSTRACT In urban traffic, a Dynamic Traffic Light System (DTLS) is an important aspect of automatic driving. DTLS estimates the time of the light signal from images of dynamically changing road traffic. In conventional traffic light systems, light signals are enabled at predefined or fixed time intervals without having information on the current traffic density on the road. This static behavior of the traffic light system increases unnecessary waiting time on the road, eventually creating traffic jams, environmental pollution, and other health emergencies. The smart traffic light system addresses these issues with self-learning algorithms and dynamically allows traffic to pass by learning current traffic density. In this paper, a vision-based DTLS is proposed using the YOLO (You Only Look Once) object detection algorithm that detects and counts the total number of vehicles on the roads of a traffic signal junction. The traffic signals are tuned based on the computed traffic to minimize the overall delay at that junction. Moreover, the traffic junctions are facilitated to communicate with the adjacent junctions to transmit the cumulative traffic delay observed. This delay is used to prioritize traffic passing through salient blocks like schools, offices, hospitals, etc. The paper aims to minimize the overhead incurred in both computations of traffic (using approximate computing) and in communication networks (using low-power technologies of IEEE 802.15.4 standard, specifically DSME MAC and/or LoRaWAN). The proposed system accomplishes its objective of smart city infrastructure by optimizing the traffic flow. Further, the paper provides a mechanism for green traffic corridors for emergency vehicles.

INDEX TERMS Edge computing, IEEE 802.15.4 DSME MAC, Internet of Things (IoT), LoRa, LoRaWAN, machine learning, YOLO.

I. INTRODUCTION

In recent years, Intelligent Transportation Systems (ITS) have emerged as a transformative technology with the potential to revolutionize the transportation sector and several Internet of Things (IoT) and Internet of Vehicles (IoV) applications [1], [2], [3]. Machine learning (ML) [4] is a subset of Artificial Intelligence (AI) that facilitates the machine to make decisions similar to humans. ML algorithms learn from the input and produce deterministic output without being explicitly programmed. ML is gaining prominence in the Internet of Things (IoT) networks [5], [6], [7]. An IoT network is a set of AI devices/sensors that communicate among devices without human intervention. The recent proliferation of IoT-based smart transportation and applications across diversified industrial domains, including the rich cloud services [8] has facilitated the need for edge computing, wherein data processing occurs in part at the network edge, rather than completely in the cloud [9], [10], [11], [12], [13]. Edge computing addresses several cloud computing

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issues, such as latency, devices' constrained power, reducing network congestion, bandwidth costs, etc. Sensors connected to edge AI devices generate a large amount of data for computation on resource-constrained edge devices [14]. It is challenging to execute ML algorithms on edge devices due to the limited memory and processing power [15], [16]. By applying AI technology to the edge, it is possible to draw fast and accurate results for analysis and decision-making in an IoT environment. By increasing the role of AI algorithms in edge computing environments, smart factories, smart cities, smart entertainment, and autonomous things can be efficiently deployed. In addition, selecting suitable algorithms, the correct dataset, and data preprocessing are other challenges in IoT applications.

Efforts across the world for ITS are focused on addressing a range of problems, including minimizing traffic accidents and fuel consumption, less congestion, reducing time spent on traffic jams, and enhancing transportation safety [17], [18], [19], [20], [21]. For example, traffic congestion systems use IoT-based consumer application devices, sensors, and machine-learning algorithms to monitor traffic patterns and adjust traffic signals in real-time. Smart traffic light system plays a vital role in both human and automatic driving. In 2017, the Insurance Institute of Highway Safety stated that around 900 people died in crashes related to faulty traffic signals in US [22]. As about 68% of the world's population will be residing in urban areas by 2050, according to a projection of world urbanization undertaken by the United Nations (UN) in 2018, the poor traffic scenarios in the cities will further deteriorate. Traditional traffic light system is based on manual configuration operating with a static time. The configuration can be set up by a professional (traffic operator) or can be controlled centrally via a wired communication network. The limitations of such a mechanism are that it does not consider the number of vehicles passing through the roads/lanes (in static timers), or a disruption in the wired connection (centralized system) can result in a failure of the traffic light system. On the other hand, a dynamic traffic light system (DTLS) avoids unnecessary waiting time on roads due to its intelligent, self-learning behavior. DTLS system decides the state of the signal by collecting real-time traffic information at any junction. Nowadays, deep ML algorithms have been popularly used for object detection to increase the efficiency of traffic light systems [23].

The IEEE 802.15.4 standard has become the most popular networking standard for IP-based IoT networks [24], [25]. The standard is suited for low-power wireless communications and defines physical layer protocols and medium access control (MAC) sub-layer specifications. The Deterministic Synchronous Multi-channel Extension (DSME) is one of the MAC modes defined in the standard that provides higher throughput with low and deterministic delay, including enhanced scalability and robustness through multi-channel access using the multi-channel DSME-GTS mechanism [26]. Therefore, it is one of the most suitable communication technology between adjacent traffic junctions. DSME facilitates low-power transmissions that enhance the decision-making process (traffic light tuning) at each traffic junction over the considered road. The MAC mode operates in beacon-enabled mode [27], where associated devices (at each traffic junction) synchronize their transmissions autonomously using the beacon frames. These beacons coordinate a DSME multi-superframe structure that defines a device's transmissions and channel access mechanism. Moreover, relay DSME devices can immediately forward data using DSME-GTS [28] with ultra-low latency. They may be installed in between traffic junctions and can additionally perform any other functions of edge computing and smart city. DSME addresses the challenges of interference caused by other wireless networks by enabling devices to coordinate and utilize channels appropriately. To achieve this, the DSME MAC protocol utilizes a predetermined hopping sequence through the channel hopping mechanism, which is repeated until data transmission is complete. Furthermore, DSME facilitates channel adaptation, enabling devices to select channels based on the identified link quality and switch channels accordingly.

Alternatively, LoRa (Long Range) technology can be used for communication between traffic junctions that are far (2)-7 km) from each other. LoRaWAN (LoRa Wide Area Network) facilitates low-power, long-range communication between devices using its chirp spread spectrum. The LoRaWAN MAC layer specifies the MAC control specifications that permit the transmission of data/messages between devices and gateways [29]. The MAC protocol provides LoRa end nodes with channel access, adaptive data rate, security, and energy-saving avenues and is highly scalable. A combination of the DSME and LoRaWAN MAC can be used depending on the distance between the traffic junctions. LoRaWAN can be explicitly used to transmit information to the base station (traffic monitoring headquarters), where the distance between them extends to around 2-7 km. In this paper, we implement a traffic system architecture with both the low-power technology co-existing and compare their performance and impact in the system.

A. MOTIVATION

In an IoT infrastructure of the smart city, effective monitoring and policing of traffic management is a crucial element of a DTLS. Traffic management policies involve real-time detection and response of traffic to ensure smooth mobility. Also, providing road safety and efficient transportation is a necessary aspect of smart traffic systems. The traditional traffic systems use pre-fixed timer circuits without having real-time scenarios of traffic in lanes. In order to mitigate the long waiting time and congestion, we need to focus on real-time information on traffic signal points. ML algorithms are popularly used to improve the efficiency of traffic signals by learning the data collected from traffic signal points.

The DTLS optimizes the duration of traffic lights depending on the inference given by the ML algorithms. Further, the necessity of providing seamless movement to emergency vehicles like ambulances, firetrucks, and organ transport vehicles across several traffic junctions needs to be addressed. This will result in the reduction of loss of life and property due to emergency vehicles being stuck in long traffic congestion.

In recent years, edge ML computing algorithms have been preferred over cloud computing due to the following three major reasons. First, processing a large amount of data on edge IoT devices saves data-transfer bandwidth. Secondly, edge computing offers more data security regarding data loss, denial of service, etc., than cloud computing [30]. Finally, edge computing has a low response and prediction time. Hence, edge computing is considered a promising solution for latency-sensitive, area and power-constrained IoT devices. In the proposed work, Raspberry Pi (Rpi) devices are used for edge devices at the traffic junction that process the input images using the YOLO (You Only Look Once) algorithm to switch/tune the traffic signals. The Rpi devices are equipped with IEEE 802.15.4 or LoRa-based transceivers that transmit the cumulative delay incurred at that junction to adjacent traffic junctions.

B. CONTRIBUTION & ORGANIZATION

This paper presents a smart traffic light system architecture that dynamically monitors the traffic condition and switches the signals accordingly. One of the primary design goals is to optimize the delay (signal waiting time) among all the lanes at each junction. Additionally, we aim to prioritize traffic passing through salient blocks like schools, offices, hospitals, etc., by computing the cumulative traffic delay along a considered road. At each junction, we have used the Convolutional Neural Network (CNN) model to infer the traffic density of each lane in a junction and process on board to make signal decisions. Also, we have developed an algorithm for the smooth and efficient movement of vehicles in traffic signal junctions. In the paper, we measure the optimized delay through DTLS by determining the waiting time for a lane between two green signals. The same metric is used to evaluate the performance of the proposed system (through Algorithm 2) in peak hours through important blocks. The primary objective is to optimize this waiting time in normal scenarios and minimize the waiting time for prioritizing traffic through important blocks.

The main contributions of the paper are summarized as follows:

- We propose a smart traffic signal system based on an optimized YOLO object detection algorithm that can classify and count the number of vehicles on the road. We have adopted structured pruning and linear quantization techniques to reduce the computational complexity of the algorithm on edge hardware.
- A smart DTLS mechanism is presented that dynamically controls the traffic signals based on the traffic congestion (number of vehicles) at each junction. It also considers the cumulative delay incurred by the vehicles on the road and prioritizes some traffic to compensate for the delay.
- An architecture facilitating low-power communication technologies of IEEE 802.15.4 DSME MAC and LoRaWAN is used for communication between the devices within a traffic junction and between different junctions, respectively.

The rest of this paper is organized as follows. The study of the existing technologies is discussed in Section II. Section IV presents the proposed dynamic traffic light system. The implementation and results are explained in Section V. Finally, the conclusions are drawn in Section VI.

II. RELATED WORKS

The ICPTS has emerged as a potential solution to address the challenges of traditional traffic light systems, which operate on fixed time intervals and do not take into account real-time traffic density. In this section, we review and analyze recent research efforts in the area of ICPTS, specifically focusing on smart traffic light systems that leverage machine learning and communication technologies to improve traffic efficiency and reduce environmental impact.

The authors in [31] aimed to improve traffic signals by developing a centralized traffic light control system, using a unique wireless communication network, termed a Smart Traffic Light Control System (STLCS). The work additionally included a pedestrian button to support people crossing the roads as well. However, it does not cater to the specific requirements of a lane. The centralized mechanism also suffers from the high overhead of control and maintenance. A synchronization mechanism among multiple traffic light systems at adjacent intersections was proposed in [32]. The authors also reserved a provision for emergency vehicles to get priority. [33] proposed a similar synchronized mechanism termed as Adaptive Traffic Light Control System (ATLCS) to assist traffic management authorities in effectively dealing with the growing traffic congestion. The primary goal of the ATLCS is to synchronize different traffic lights at consecutive junctions by creating a delay between the times at which each of them switches to green in a given direction. The delay can be dynamically updated based on the count of cars waiting at each junction. Although both two systems are successful in maintaining steady traffic in a single lane, they do not consider the probable congestion in other lanes in the given junctions. On the other hand, Lee et al. [34] tried to develop a smart traffic signal control (STSC) system that includes several smart city transportation applications, including emergency vehicle signal preemption (EVSP), public transport signal priority (TSP), adaptive traffic signal control (ATSC), eco-driving supporting, and message broadcasting. The system also facilitates informing the drivers of any emergency vehicle approaches, such as ambulances, fire trucks, etc. However, the system is complicated, as it requires several wireless and cellular technologies such as 3G/4G, Wifi/Bluetooth, and cloud services. In [35], the authors aim to measure traffic

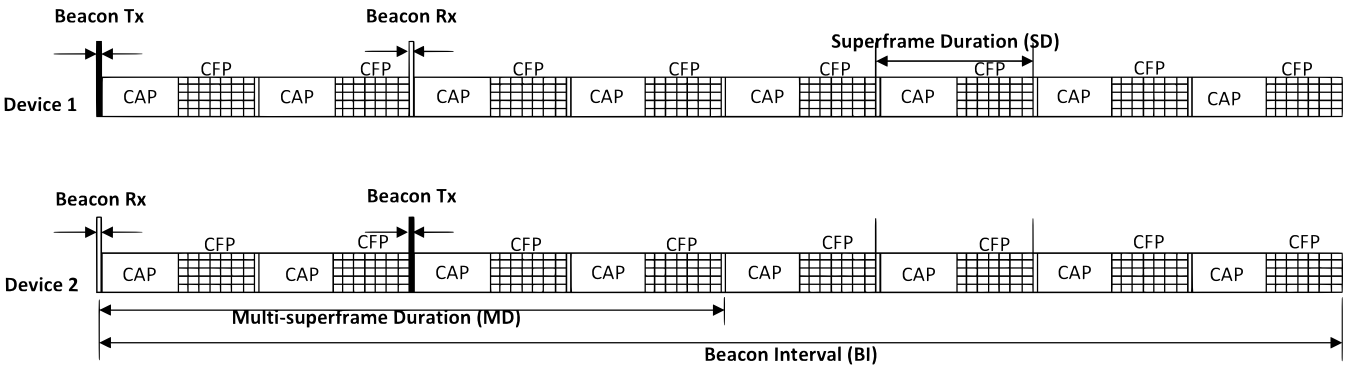


FIGURE 1. DSME multi-superframe structure.

density with the aid of IR sensors. Several IR sensors are placed in each of the lanes that measure the traffic density at each of the lanes, and based on this information, the traffic lights are dynamically changed. A similar solution was provided by [36] using LoRaWAN technology. However, the aforementioned schemes fail to maintain a uniform delay (through traffic lights) incurred by vehicles on a particular road. That is, a road (deemed important) consisting of schools, offices, and hospitals as destinations will have heavy traffic compared to other adjacent roads in the city. Measuring the traffic with IR sensors or other wireless technologies will result in long traffic delays on that considered road as that road will have been identified as the most congested road.

The traffic control system developed by Megalingam et al. [37] relied on information transmitted by wireless sensor network (WSN) nodes to facilitate optimal traffic routing and monitor high-speed vehicles. It's worth noting that the authors did not execute a practical implementation of the system but rather created a platform for data exchange between WSN nodes and a central monitoring station (CMS). Cunha et al. [38] have proposed various wireless communication systems for traffic control and management, as documented in several works [38], [39], [40]. The author utilized ZigBee as a communication protocol for transmitting information between traffic lights. In contrast, a long-range communication protocol (LoRa technology) was employed for communication between intersections and the management center. While the system provided several operational calculations for traffic lights based on the number of vehicles crossing, it did not include vehicle detection sensors. Instead, the calculations were based on random and deterministic values. Additionally, the system addressed only one intersection type. Given the abundance of information available on traffic congestion through various means (sensors, adjacent traffic systems), the use of machine learning has the potential to transform the decision-making process. A system for the passage of emergency vehicles termed STLS (Smart Traffic Lights System) was proposed in [41]. The proposed system uses an app along with Google Maps that allows the emergency vehicle to send information about its arrival time at each traffic signal junction. Eventually, when the vehicle arrives at the traffic lights, it will find them open without any conflict with the other signals.

The authors in [42] propose a reinforcement learning-based approach to update the traffic lights dynamically. Similarly, a cognitive-based road traffic management system was proposed in [43]. A similar approach was presented in [44]. Liang et al. [45] proposed a control mechanism for the traffic light cycle based on Deep Reinforcement learning. The model represents the traffic scenario as states, and the intersection is divided into blocks. Markov decision process is used to model the traffic light duration change. The main limitation of this system is the release of one lane that is not dependent on traffic from other lanes. Also, in the aforementioned schemes [41], [42], [43], [44], [45], the computational overhead arising from the learning methodologies are observed to be very high, and the data generated by the traffic junctions are transmitted to a centralized system and incurs delay in the decision process.

In view of the above limitations, we propose a lightweight DTLS mechanism using the YOLO object detection algorithm based on which the traffic lights are adapted in order to minimize the overall traffic delay across all the lanes in a traffic junction. Additionally, to compensate for the traffic delay incurred on important roads of a city, a priority system is developed to compute the cumulative delay at each of the traffic junctions on that road and compare it with other adjoining roads/lanes of the considered road.

III. OVERVIEW OF IEEE 802.15.4 DSME AND LORA

A. IEEE 802.15.4 DSME

DSME is designed to cater to the demands of applications that have strict Quality of Service (QoS) requirements. This is achieved by employing techniques such as channel hopping and channel adaptation. DSME ensures that multiple channels are available for all transmissions during the Contention-Free Period (CFP). The enhancement of multi-channel access can be achieved through the application of the CAP reduction technique, as mentioned in [27].

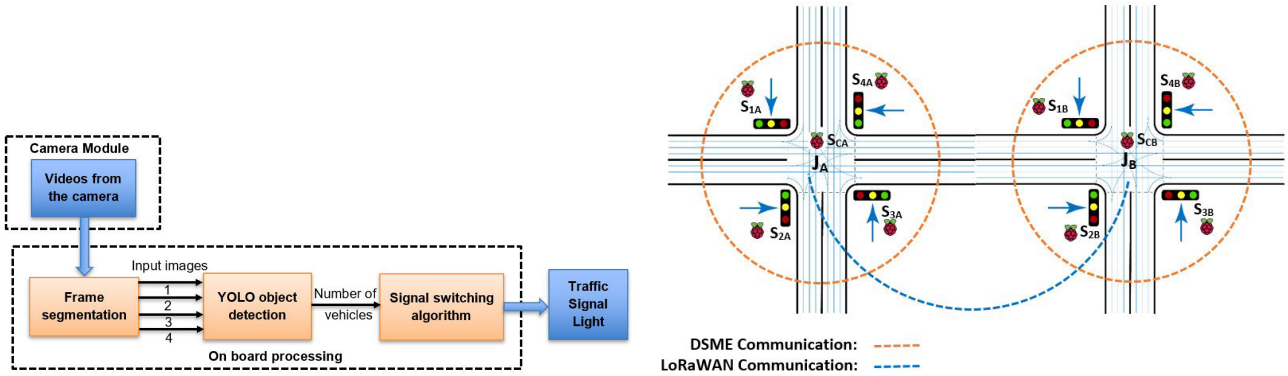


FIGURE 2. Proposed smart traffic light system. (a) Block diagram; (b) Representation in a real-world scenario.

When CAP reduction is implemented, all CAPs except the one in the initial superframe of the multi-superframe are replaced with DSME-GTSs [27]. To address issues stemming from interference caused by other wireless networks, DSME enforces effective coordination among devices for managing channel usage. Within the DSME MAC protocol, a channel hopping mechanism is integrated, featuring a predefined hopping sequence that persists throughout the entire data transmission process. In contrast, DSME incorporates channel adaptation, allowing devices to autonomously change channels in response to the evaluated link quality.

In a DSME network, devices coordinate their transmissions within a time framework referred to as a "multi-superframe," which is an extension of the IEEE 802.15.4 superframe structure [46]. This multi-superframe comprises a total of sixteen timeslots, each of equal duration. The first slot within this structure is dedicated to beacon transmissions, while the subsequent eight slots are allocated for transmissions using the CSMA/CA mechanisms. This transmission phase is known as the "Contention Access Period" (CAP). Following the CAP, there is the "Contention-Free Period" (CFP), during which DSME-GTSs are utilized across multiple channels for transmissions. The combined duration of the transmission period, including both CAP and CFP, is referred to as the "Superframe Duration" (SD). Additionally, the "Beacon Interval" (BI) parameter is defined as the time span between successive beacon transmissions. The BI encompasses multiple multi-superframes, each of which contains numerous superframes. A novel parameter introduced in this context is known as "multi-superframeOrder" (MO), which signifies the duration of the entire multi-superframe, encompassing all individual superframes within it. Two other essential superframe parameters, "macBeaconOrder" (BO) and "macSuperframeOrder" (SO), jointly define the structure of the multi-superframe, as illustrated in Fig. 1. These parameters are interconnected by the following constraints: $0 \leq SO \leq MO \leq BO \leq 14$. The values for BI, SD, and MD are determined using the following mathematical expressions:

$$\mathrm{BI} = \mathrm{aBaseSuperframeDuration} \times 2^{\mathrm{BO}} \tag{1}$$

$$\mathrm{SD} = \mathrm{aBaseSuperframeDuration} \times 2^{\mathrm{SO}}, \tag{2}$$

$$\mathrm{MD} = \mathrm{aBaseSuperframeDuration} \times 2^{\mathrm{MO}}, \tag{3}$$

The term "aBaseSuperframeDuration" is defined as the quantity of symbols that make up a superframe when the SO is configured to zero. Within the Beacon Interval (BI), there are $2^{\mathrm{BO-SO}}$ superframes and $2^{\mathrm{BO-MO}}$ multi-superframes. During network initialization, the PAN coordinator establishes the parameter values for BO, SO, and MO, and these values are disseminated to associated devices through the use of Enhanced Beacons (EB) [27]. The EB plays a crucial role in maintaining network synchronization. Referring to Fig. 1, it becomes evident that a device transmits a beacon during its designated Beacon Tx slots, while it receives beacons from another coordinator during the Beacon Rx slots. In Figure 1, initially, Device 1 transmits a beacon in its Tx slot, and simultaneously, Device 2 receives this beacon in its Rx slot. At a later point in time, Device 2 takes its turn to transmit a beacon in its designated Beacon Tx slot, and Device 1 receives the transmitted beacon in its respective Beacon Rx slot.

B. LORA

LoRa, as discussed in [47], is a low-power wide area network (LPWAN) technology designed to enable long-distance communication for IoT devices. It serves as an ideal wireless communication solution for various scenarios, including both urban and rural environments, where low data rates, extended battery life, and long-range connectivity are essential. LoRa technology, based on the chirp spread spectrum (CSS), is renowned for its reliable communication, particularly in congested areas. By utilizing a wide frequency band to spread signals across the electromagnetic spectrum, it excels in long-distance communication and is capable of covering vast distances in open spaces. Its low-power design promotes energy efficiency, allowing IoT devices to operate for

years on a single battery charge. LoRa employs a star-of-stars network topology, making it suitable for large-scale deployments.

LoRaWAN, or Long Range Wide Area Network, represents a wireless communication technology tailored for IoT applications. It specializes in providing extended-range connectivity for low-power devices, making it well-suited for applications like smart agriculture, asset tracking, and environmental monitoring. LoRaWAN boasts extended coverage, efficient battery utilization, and secure data transmission, rendering it a popular choice for IoT implementations. It operates on various frequency bands, supports bidirectional communication, and adheres to open standards, facilitating interoperability among diverse devices and infrastructure. Its adaptability and scalability position it as a key player in the IoT ecosystem.

The LoRa network comprises five key components: end nodes (also known as end devices), gateways, the Network Server (NS), the Join Server (JS), and application servers, all interconnected in a star topology architecture. The LoRa end node consists of a wireless transceiver and sensor nodes that transmit data to multiple nearby gateways using LoRa radio frequency (RF) modulation. Gateways, powered by mains and equipped with internet connectivity, incorporate both a radio component for transmission and a microprocessor for data processing. Each gateway in the network forwards received data packets to the cloud-based NS, which subsequently routes the packet to the appropriate application server. In scenarios with multiple gateways, all of them can receive data from the same end node. Gateways can simultaneously monitor multiple frequencies within each Spreading Factor (SF).

C. COMPARISON

LoRa and IEEE 802.15.4 are low-power wireless networking technologies that offer several benefits. LoRa can cover several kilometers in rural areas and a few hundred meters in urban areas. The LoRa devices can run on batteries for up to 10 years or more, making them suitable consumer devices. Further, LoRa radio modulation provides deep indoor coverage, connecting end devices located in hard-to-reach areas. LoRa networks can scale well, thanks to the possibility to use multiple gateways and be equipped with different radios, capable of receiving signals transmitted with different spreading factors at the same time. This makes LoRa a suitable solution for communicating the cumulative delay from one traffic junction to another. Similarly, IEEE 802.15.4 is designed for low-power, low-data-rate applications, making it ideal for devices that need to run on batteries for a long time. Additionally, IEEE 802.15.4 can coexist and interoperate with existing network architectures. However, the communication range is lower compared to LoRa and consumes lower power. Hence, it is suitable for the transmission of signal information and vehicle count between the different devices (deployed in the lanes) within a traffic junction. Both of the low-power wireless networks are preferred over WiFi and Bluetooth for IoT applications that require long-range and low-power communication. While Bluetooth is ideal for devices in close proximity, LoRa's long-range communication and low-power consumption make it a better choice for our smart traffic IoT applications that require wide-area coverage.

IV. PROPOSED SYSTEM DESIGN

In this work, a smart DTLS is designed for multi-direction, multi-lane traffic junctions. The video camera will continuously record the density of traffic coming from each road in a junction. In this system, we are considering double-lane roads, where cameras are installed on the lane where vehicles are approaching the junction. We have processed the data collected from the camera on board with the help of the YOLOv3 model. Fig. 2(a) illustrates the basic block diagram of the proposed DTLS system. In this figure, the camera module contains four video cameras that are installed separately in four different lanes of a junction to capture the vehicle density. S_{1A}-S_{4A} and S_{1B}-S_{4B} are the camera modules installed in two adjacent traffic junctions J_A and J_B, respectively as shown in fig. 2(b). Videos are segmented in image frames at regular intervals of time and fed to the object detection module to count the number of vehicles. The proposed switching algorithm decides the traffic light signal. Furthermore, an average delay at each junction (across different lanes) is computed and transmitted to the adjacent junction. At the next junction, the received average delay is added to its own average delay, and the cumulative delay is transmitted toward the next adjacent delay. The cumulative delay data can be transmitted via cloud computing using an MQTT service; however, in this work, we refrain from using cloud services due to the disadvantages mentioned in Section I. Therefore, we perform the necessary computations at each device at the traffic junction (edge computing) and transmit these data to the neighboring device (relay or junction device).

The proposed architecture employs two popular low-power, low-cost link layer communication protocols, namely IEEE 802.15.4 DSME and LoRaWAN. The DSME operates among the IoT devices within a junction deployed at the different lanes, as shown in Fig. 2(b). For example, devices S_{1A}, S_{2A}, S_{3A}, S_{4A} and the central computing edge device, S_{CA} in junction, J_A communicate with each other using IEEE 802.15.4 DSME. This is because the communication range of DSME is around 70-100 meters, and a data rate of up to 250 kbps can be achieved. This ensures faster communication between the devices so that the requisite computing and signal can be transmitted. DSME-GTS [28] slots are used by each device to guarantee transmission without fail and contention. On the other hand, LoRaWAN link-layer communication technology is used for communicating the computed cumulative delay from one traffic junction to the other. This is shown in Fig. 2(b), wherein the central edge device, S_{CA} transmits the delay to the device, S_{CB} at the adjacent junction, J_B. Since LoRa devices can transmit up

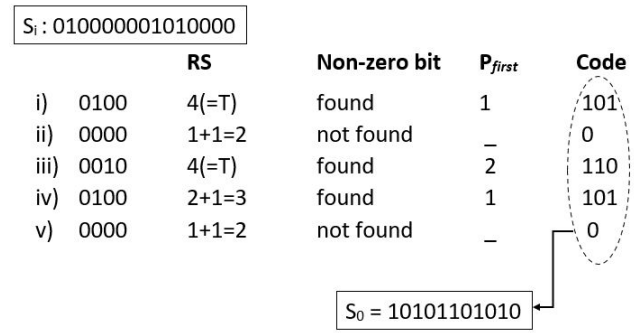


FIGURE 3. Illustrative example of RAKE algorithm.

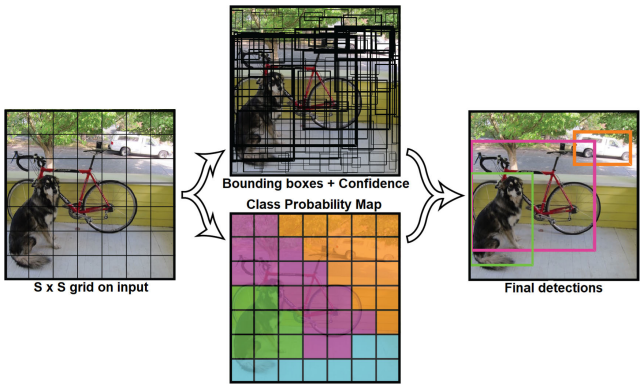


FIGURE 4. Grid division and detection of objects using YOLO (adopted from [49]).

to the range of 2-7 km [29], the traffic junctions could be deployed far away from each other.

The detailed functioning of the proposed DTLS is described in the following subsections.

A. IMAGE APPROXIMATION

Image approximation is crucial for IoT applications to reduce bandwidth requirements, storage resources, and power consumption of battery-powered edge devices. Image compression algorithms can be used in image approximation to design a cost-effective IoT system. Although lossy compression algorithms offer a much higher compression ratio than lossless compression algorithms, the latter is more widespread in object detection applications to maintain detection accuracy. In lossless image compression algorithms, a trade-off between compression ratio and computational complexity (i.e., power, area, and accuracy) is considered for resource-constrained edge computing scenarios. In this work, RAKE (Random Access and K-Lossless Encoding) lossless compression algorithm [48] is used due to its low complexity, which efficiently encodes positions of non-zero bits. Moreover, the RAKE algorithm achieves a higher compression ratio in real-time data and is suitable for devices having limited storage and computational resources. Before image compression, we pre-processed the RGB input images into grayscale images. Further, in many cases, grayscale images preserve sufficient shape and texture information from their original RGB representation to describe classes of objects. Finally, 16-bit grayscale images are considered as input to the RAKE compression algorithm. Fig. 3 illustrates an example of RAKE compression algorithm with 15-bit input binary data $S_i = 010000001010000$ and sliding window (rake) of length $T = 4$. The number of bits shifted by the rake is calculated as $RS = P_{first} + 1$, whereas P_{first} is the position of the non-zero bit in the T-rake. A code word is generated, and operations continue until the rake reaches the end of the input data sequence. The compressed output S_o of 11-bit is generated at the end of the execution by concatenating all the code words.

B. YOLO OBJECT DETECTION ALGORITHM

YOLO [49] network only looks at the image in one go to detect multiple objects. The incredibly fast processing speed and small size make YOLO advantageous for different object detection applications. The YOLO algorithm identifies the part of the image having a high probability of containing an object and predicts bounding boxes with class probabilities of detected objects. It requires only one forward propagation pass through the CNN to make predictions. Fig. 4 illustrates how YOLO divides an input image into a grid division. In this work, tiny-YOLOv3 [50] has been selected as it runs in real-time and outperforms other state-of-the-art deep learning object detection algorithms like Faster R-CNN [51] and SSD (Single Shot Detector) [52] in accuracy. DarkNet model is the backbone of the YOLOv3 model and it can process a maximum of 45 FPS (Frame per second), however, tiny-YOLOv3 can process up to 220 FPS. Video streams captured by the cameras are preprocessed by using functions from OpenCV libraries before being sent to the YOLO detection model. Preprocessing involves converting video streams to frames with a predefined frame rate and resizing the frame or input image. YOLO works with pre-trained weight files, model configuration files, and text files containing classes of the objects to be detected in an input image. We have used an optimized tiny-YOLOv3 model on the TensorFlow framework to count the number of vehicles in real time from the video stream of a traffic signal. The tiny-YOLOv3 model is pre-trained on the Common Object in Context (COCO) dataset [53] for vehicle detection and classification. The proposed switching algorithm takes the signal decision based on the vehicle counts on each lane.

1) OPTIMIZED YOLO

Although YOLO has several updated versions based on model compression, deploying on-edge devices for real-time computing is still very challenging. The tiny-YOLOv3 has excellent potential for designing compact, low-power circuits for edge-embedded devices where real-time processing or deployment on resource-constrained devices is crucial. Tiny YOLOv3 is a smaller and faster version of YOLOv3 that sacrifices some accuracy for increased speed and reduced computational requirements. In practice, compressing the network structure is very crucial for application-specific

hardware designs. The compression strategy involves the removal of some convolutional layers, output layers, and redundant parameters. In this work, we have optimized the existing tiny-YOLOv3 model by pruning during the training to cut out some weights or unimportant connections that do not affect and contribute to the final result. Pruning reduces the inference time and volume of the YOLO network parameters resulting in the reduction of storage space. The structured pruning [54] technique is adopted to offer efficient processing in hardware by exploiting the data-parallel architecture. In addition, we quantize the tiny-YOLOv3 network during inference by using linear transformation once at the beginning of the network. Quantization changes quantized levels of the parameters and activation functions into low-precision to avoid costly FLOPs (Floating Point Operations). Processing of FLOPs on hardware is computationally more intensive and complex than integer operations. The transformation from floating point to 8-bit integer is calculated as:

$$x_q = int\left(x \times scale\right) \tag{4}$$

where, $scale = min\left(abs\left(\frac{127}{x_{max}}\right), abs\left(\frac{-127}{x_{min}}\right)\right)$

x_q is the n-bit quantized integer of the floating point number x. Similarly, dequantization is calculated at the end of the network as follows:

$$x = int\left(\frac{x_q}{scale'}\right) \tag{5}$$

where, $scale' = min\left(abs\left(\frac{2^{n-1}-1}{x_{max}}\right), abs\left(\frac{-2^{n-1}-1}{x_{min}}\right)\right)$

The quantization and dequantization accelerate the computation of network layers, making them cost-effective for hardware implementation. The performance of the optimized tiny-YOLOv3 algorithm is compared with the original tiny YOLOv3 and YOLOv3 in terms of mAP (mean average precision) and FLOPs on the COCO dataset. Experiment on the dataset shows that the optimized tiny-YOLOv3 reduces the amount of FLOPs while maintaining decent detection accuracy.

C. PROPOSED TRAFFIC LIGHT SYSTEM

In this paper, we propose a dynamic traffic light system with a switching mechanism that is based on the shortest job first (SJF) and round-robin algorithm. The proposed mechanism initially uses the optimized YOLO mechanism to compute the number of vehicles waiting in each lane. The traffic light is turned green in the lane with the least number of cars. This is done to avoid a large waiting time. It is known that implementing SJF technique gives the least overall waiting time. However, the road (or lane) with a larger number of vehicles can suffer from starvation of green signals if only the road with the smallest count of vehicles is given priority. Hence, we implement the round-robin approach among all the traffic signal directions to address the problem. The dynamic traffic light system algorithm is shown in Algorithm 1.

Initially, the edge devices along the lanes of the traffic junction capture images of the vehicles. Based on the image that is set for onboard processing detects the number of vehicles. The proposed algorithm counts the number of vehicles across all the available lanes at the junction. This is done by transmitting the vehicle count from each of the lanes to the main central edge device at the traffic signal junction. This transmission is achieved through wireless communication as the edge devices equipped with camera modules are generally located within 1-30 meters apart. The IEEE 802.15.4 compliant transceivers would be a perfect choice for this implementation. In this paper, we use the IEEE 802.15.4 DSME MAC module for transmitting vehicle counts. Next, the algorithm sums and computes the fraction of vehicles in each of the lanes, sorting them in increasing order of vehicle count. That is, the lane with the least number of vehicles is chosen to flow the traffic first. The signal timing for each lane is dependent on its respective fraction of vehicles at that lane. This ensures an even waiting time at each lane.

Algorithm 1 Proposed Dynamic Traffic Light System

Retrieve the number of vehicles, $lane_{vehicle-count_i}$ for each of the signal lane i from n available lanes ;
Compute $N = \sum_i^n lane_{vehicle-count_i}$;
Compute the fraction of vehicle at each lane, $F_{lane_i} = \frac{lane_{vehicle-count_i}}{N} \times 100\%$;
Sort the signal lanes i in increasing order of vehicle count in a list $\mathcal{B}[n]$;
for *each* $i \in \mathcal{B}$ **do**
  Activate the green traffic signal based on the list $\mathcal{B}[n]$;
  Continue the signal for F_{lane_i} time units.
end

Once the green signaling time for a lane has exceeded, the next lane, as per the $\mathcal{B}$, is allowed to flow. This would be repeated for all the lanes in $\mathcal{B}$. Once completed, the Algorithm 1 would be invoked to compute the next set of $\mathcal{B}$.

D. TRAFFIC PRIORITY USING CUMULATIVE DELAY

To achieve priority-based signaling, each traffic junction should be aware of the prior delays incurred through the route. This is facilitated through the transmission of the cumulative delay from one junction to another. A road trip is considered (that contains important offices, schools, buildings, etc.) wherein the deployed device (at the first junction) keeps a count of the delay incurred in waiting between other signals at the crossing. This delay is transmitted to the next device (at the next junction) through LoRa communication in its forward direction of traffic, as shown earlier in Fig. 2(b). At the traffic junction, the cumulative delay is computed, i.e., the delay incurred at that point and the delay received from the previous junction. This cumulative delay is used to make and overwrite signal decisions at that junction. This could be implemented during peak office/school hours or any

particular time. As per Algorithm 1, the signal is adapted based on the number of vehicles; however, with the priority mechanism, the cumulative delay is the primary parameter considered along with the vehicle count. The objective is to have minimal overall delay along the route. This is illustrated in Algorithm 2. This can be executed in all the traffic directions on the route.

Algorithm 2 Traffic Priority With Cumulative Delay

Set Cumulative delay, $Delay_{cum}$ =0;
for *each* $junction_i \in road_{strip}$ **do**
  Compute the delay incurred, $delay_i$;
  Transmit $delay_i$ to the next $junction_{i+1}$;
  $Delay_{cum} = Delay_{cum} + delay_i$;
end

E. SMART GREEN CORRIDOR

The necessity for the smart green corridor has increased in recent times, especially in urban metro cities. The traffic congestion observed in such cities results in several mishaps that could have been prevented. Emergency services like ambulances, firetrucks, and organ transportation require complete traffic-free movement. Currently, this is done manually by the traffic personnel and only upon pre-informed. Additionally, these emergency services get stuck in traffic congestion, resulting in loss of life, property, etc. To avoid this, we propose a Smart Green Corridor (SGC) mechanism that assists emergency vehicles in traveling through traffic junctions without any halt, i.e., provides green signals at each of the junctions they pass through.

Algorithm 3 Proposed Smart Green Corridor

Identify emergency vehicle at a traffic junction, T_{junc_i};
Set $\mathcal{E}_i$=1;
Activate the green traffic signal;
for *each* $\mathcal{E}_i$*=1 in the adjacent Junction* **do**
  Transmit $\mathcal{E}_i$=1 to $T_{junc_{(i+1)}}$;
  if T_{junc_i}*==1(red signal)* **then**
    Overwrite & activate the green traffic signal;
  end
  else
    Continue with the green traffic signal;
  end
end

The cameras at the junctions identify emergency vehicles. This information is passed from one traffic junction to another using IEEE 802.15.4 DSME or LoRa communication technology, or a combination of both. Once a traffic junction is aware of an upcoming emergency vehicle through it, the signals are kept green on the lane through which the emergency vehicle is supposed to pass. This may necessarily overwrite the signal, supposed to be tuned as per Algorithm 1, as it is of higher priority. The SGC algorithm is shown in Algorithm 3. The proposed DTLS algorithm is combined with the proposed Smart Green Corridor (SGC) algorithm.

TABLE 1. Performance comparison of proposed optimized Tiny-YOLOv3 algorithm.

Parameters	YOLOv3	Tiny-YOLOv3	Optimized Tiny-YOLOv3
Model Size (MB)	132	23.8	*20.1*
mAP (%)	89.73	80.26	*81.93*
BFLOPs	34.29	5.46	*5.02*

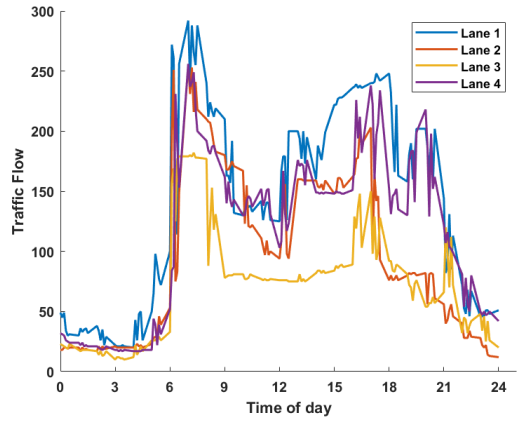


FIGURE 5. Traffic flow per lane on a day.

In this work, video streams are preprocessed by using functions from OpenCV libraries before being sent to the tiny-YOLOv3 detection model. Preprocessing involves converting input video streams to frames with a predefined frame rate and resizing the frame or input image. Frame rates depend on traffic density or the number of vehicles at any given time. Resizing reduces the computation complexity in terms of time and resource consumption. Once YOLOv3 computed the number of vehicles and sent the counts to Algorithm 1, it would be invoked again after completing a cycle of green signaling for all the lanes. It is dynamic in nature and helps in making optimal traffic signaling decisions even in peak hours. Vehicle count impacts the decision of signaling time as the duration of the green signal is based on the fraction of vehicles in that particular lane (given in Algorithm 1).

V. EXPERIMENTAL RESULTS

In this section, we perform experimental analysis to measure the performance and efficiency of the proposed mechanisms using both simulations and experiments in a real test bed. The performance of the proposed DTLS mechanism is based on the

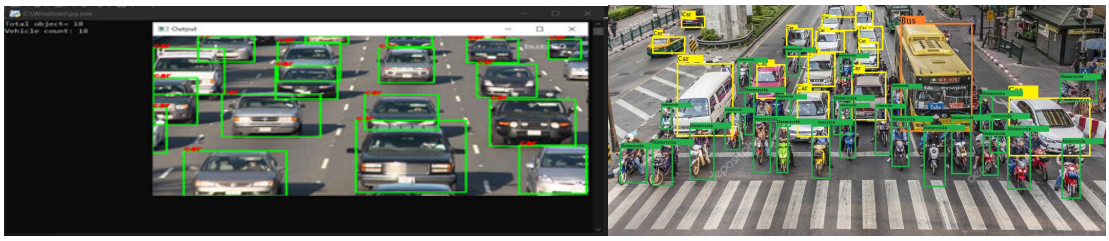
A. SIMULATION RESULTS

1) YOLO PERFORMANCE

The COCO dataset used in this work consists of 80 different classes with 3,30,000 images. The relevant classes available

(a) (b) (c) (d)

FIGURE 6. Detection and classification of vehicles.



(a) (b)

FIGURE 7. Vehicles classification and count on the road.

in the COCO dataset for this work are bicycle, car, motorcycle, bus, and truck. For the emergency classes of vehicles, including Ambulances and Firetrucks, we have collected approximately 1200 images from Google search. In addition, mAP, model size, and Billion Floating Point Operations (BFLOPs) metrics are calculated to evaluate the performance of the optimized tiny-YOLOv3 model. The mAP of all classes (seven classes in this work) is used to evaluate the accuracy of the detection algorithm and BFLOPs signify the computational power of the network model, primarily in terms of the overall amount of calculation. Table 1 demonstrates the comparative analysis of the performance of YOLOv3, tiny-YOLOv3, and optimized tiny-YOLOv3 algorithms on the COCO dataset. The optimized tiny-YOLOv3 algorithm achieves better detection accuracy with reduced BFLOPs than the original, making it suitable for real-time applications and deployment on devices with limited computational capabilities.

We validate the working of the proposed YOLO model by conducting a detection experiment using the Google Colaboratory execution environment. Fig. 6 demonstrates the efficacy of the proposed optimized object detection model that successfully detects and classifies the vehicle from any random input video stream. The count of vehicles and classes of vehicles on traffic signals detected by the proposed model are demonstrated in Fig. 7(a) and Fig. 7(b), respectively. The test images used in this illustration are publicly available images/video streams. In vision-based dynamic traffic signal, choosing optimal locations for the cameras have a significant role in clean data acquisition that affects the efficiency of the system. The camera modules are placed and configured to capture videos so as to capture maximum distance (maximum possible cars). The accurate distance for which vehicles are counted would depend on the camera specifications. However, as we have considered the COCO dataset for

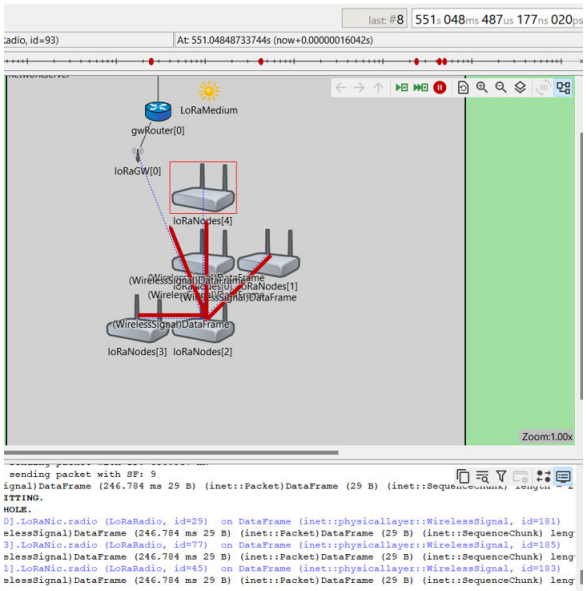


FIGURE 8. Simulation in omnet++.

training and testing of the detection model, we do not employ any camera-based data acquisition mechanism in this paper. Further, our optimized YOLO model is implemented for the real-time scenarios of traffic junctions to count the number of vehicles passing through every lane at a certain period. A graphical representation of the number of vehicles on different lanes for a day is shown in Fig. 5. These graphs can be used for traffic flow prediction in the future.

2) NETWORK DESIGN

The Network Simulator OMNeT++ [55] is used to evaluate the performance of the proposed schemes. OpenDSME [56] and FLORA modules [57] are used to implement the IEEE

TABLE 2. Simulation parameters.

Parameters	Values
Duty Cycle	0.01
Packet Length	20 bytes
Preamble Length	8 bytes
Frequency	868 MHz
Bandwidth	125 KHz
Number of nodes	60
Coding rate	25 kbps
Spreading Factor (SF)	{7,8,9,10,11,12}
Transmitting Power (TP)	{2,5,8,11,14, 17, 20} dBm
Beacon order	8
Superframe order	2
Multisuperframe order	2
Number of channels	12
Frame Length	75 Bytes

802.15.4 DSME MAC mode and LoRa in OMNeT++. The network topology considered is shown in Fig. 8. The parameters that are used in the experiments are listed in Table 2. The 868 MHz frequency band is chosen for transmissions. The Spreading Factor (SF) and Transmitting Power (TP) parameters are used for LoRaWAN communication. A larger SF increases the time on air, which increases energy consumption, reduces the data rate, and improves communication range. The power level (P) of a LoRa radio can be modified within a range of -4 dBm to 20 dBm, with adjustments made in 1 dB increments. However, due to constraints in hardware implementation, the practical range is usually restricted to 2 dBm to 20 dBm. The simulation duration is fixed at 3000 seconds duration. For any given combination of simulation parameters, we ran 30 different simulations and finally averaged over all the 30 different results. The performance metrics considered are 1) network overhead, 2) power consumption, 3) latency, and 4) throughput.

i. *Network Overhead*: The design goal of any mechanism to be implemented in resource-constrained IoT devices is to be lightweight and not incur high overhead (both control frames and data frames). The proposed algorithms have minimal computational needs and transmit only the cumulative delay (and the $\mathcal{E}_i$ for the emergency vehicle, if any) from one junction to another using the LoRa, which in itself is a long-range low-power and low-processing link-layer communication technology. Communication between the devices in a junction to determine the lane to allow traffic flow is accomplished using IEEE 802.15.4 technology. The transmission overhead of Algorithm 1 is merely the collection of information between the IoT devices in the junction. For Algorithm 2 and Algorithm 3, the overhead is the transmission of the cumulative delay, $Delay_{cum}$, and $\mathcal{E}_i$ value from one junction device to another. The transmission overhead for all the related schemes is shown in Fig. 9. It can be observed that STLS [41] has a similar transmission overhead compared to the proposed mechanism. STLS focuses mainly on adapting traffic signals for emergency vehicles. In the proposed mechanism, the IoT devices in the junction only transmit once to the main edge computing device, which in turn carries out the necessary computations and adapts the signal. In the proposed mechanism, only the computed cumulative delay is transmitted to the next junction. However, the other schemes [31], [33], [34], [41] suffer from higher transmissions due to the exchange of multiple control messages between the various devices involved in the system.

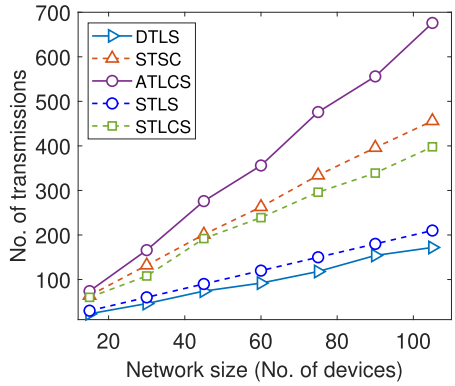


FIGURE 9. Network overhead.

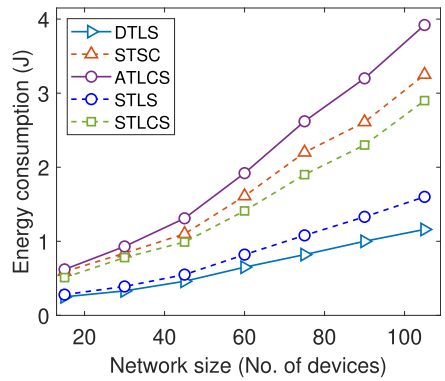


FIGURE 10. Power consumption.

ii. *Power Consumption:* As battery life is an important parameter for IoT devices, it is of utmost importance that the link layer technologies do not incur significant power while transmitting data. As power consumption is directly proportional to the transmissions, the proposed mechanism permits a longer network lifetime. Firstly, the approximate computing technique implemented in the proposed system assists in reducing power consumption. Secondly, the lightweight nature of the Algorithm 1 further results in lower power consumption, as shown in Fig. 10. It can be observed that the proposed DTLS mechanism has the least power consumption compared to the other related schemes. Higher power consumption is detected for schemes like ATLCS, STSC, and STLCS as the devices use power-hungry enabling technologies such as WiFi, Bluetooth, 4G/5G, and GPS connectivity.

iii. *Latency:* It is imperative for the network to provide low and deterministic latency. The proposed system

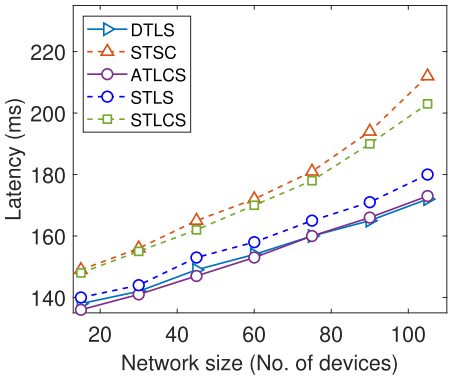


FIGURE 11. Latency comparison.

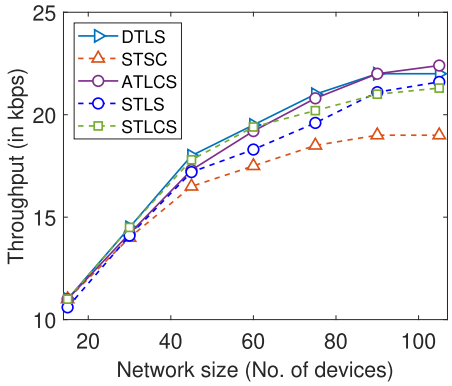


FIGURE 12. Throughput comparison.

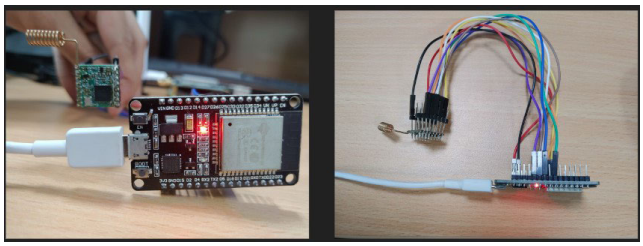


FIGURE 13. LoRa end-device: front (left) and top (right) views.

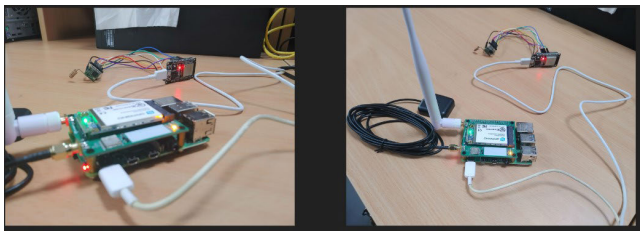


FIGURE 14. LoRa Gateway: front (left) and top (right) views.

assists the network in transmitting information with minimal delay. DSME uses GTS [27] to transmit data that is contention free in nature, thereby incurring a minimal delay. Similarly, LoRa uses different spreading factors to diversify the transmissions and minimize collisions. ATLCS and the proposed DTLS scheme have the least latency in transmitting the information. ATLCS primarily focuses on minimizing the delay of long-distance vehicles through the synchronization of traffic lights using mathematical models. However, these computations result in higher power consumption, as observed earlier. On the other hand, DTLS, with the benefits of both DSME GTS slots and LoRa communication, is ideal for real-time and adaptive traffic signal systems.

iv. *Throughput:* It is an essential QoS parameter that provides a measure of the successfully transmitted data per unit of time. As discussed earlier, the use of DSME GTS and diversified spreading factors in the enabling technologies ensures that packets are successfully transmitted with minimal contention in the channel. This results in slightly improved throughput in the network compared to other related schemes, as shown in Fig. 12. It can be observed even on scaling up the network size, ATLCS has a similar throughput as DTLS and is slightly higher than STLS and STLCS. This can be attributed to the synchronization of the traffic signal junctions.

v. *Delay:* The performance of the proposed system is compared with the state-of-the-art works and presented in Table 3. The maximum time delay (MTD) of a road and the average time delay (ATD) of all four roads are considered as metrics to evaluate the performance of the proposed system. The experiment was done on input video at different time frames of a day for one hour and measured the system's performance for different traffic densities.

B. IMPLEMENTATION OF THE SMART TRAFFIC SYSTEM IN A TEST-BED SETUP

The proof-of-concept of the proposed DTLS mechanism implementation was realized with a small testbed setup. For the IEEE 802.15.4 DSME network setup, Raspberry Pi 3 (RPi) and Openlabs RPi 802.15.4 radio are used. Openlabs offers an antenna add-on module that turns an RPi into a 6LoWPAN border router solution. The module consists of a chip antenna, a crystal oscillator, and an Atmel AT86RF233, which includes the transceiver to get an 802.15.4 solution. It provides an IEEE 802.15.4-compliant physical layer that adopts the 2.4-GHz PHY layer with a data rate of 250 kb/s. The module can be plugged directly into pins 15-26 of the RPi's GPIOs. We use a 10,000 mAh battery to power the RPi. In contrast, for the LoRa network setup, we used an ESP-32 microcontroller with an RFM95W radio transmitter that operates on 868MHz to act as our end device. For the gateway, Dragino PG 1302 Concentrator on RPi 4 was used. The concentrator aggregates the LoRa signals, and Raspberry Pi sends the data to the network server. We computed the throughput of the proposed system in the testbed created and observed that the throughput is somewhat constant between 13.5 kbps to 14 kbps, as shown in Fig. 16. Note that the simulation parameters for the experiments were considered according to the hardware parameters. The results

TABLE 3. Performance comparison of the proposed system.

	MTD (Sec)					ATD (Sec)				
Time of a day (hr)	6.00	10.00	14.00	18.00	22.00	6.00	10.00	14.00	18.00	22.00
DTLS	2.65	4.64	3.87	9.21	6.21	1.21	3.29	3.02	5.14	4.69
STSC	6.96	17.08	12.11	24.01	13.07	4.91	10.17	9.98	19.16	10.02
ATLCS	6.22	11.49	10.65	16.77	15.23	4.02	9.64	8.37	12.64	11.85
STLS	7.31	20.12	16.37	27.32	16.02	5.28	13.06	12.54	24.19	12.30
STLCS	3.48	9.27	7.58	16.07	11.83	3.01	5.35	9.14	10.35	4.71

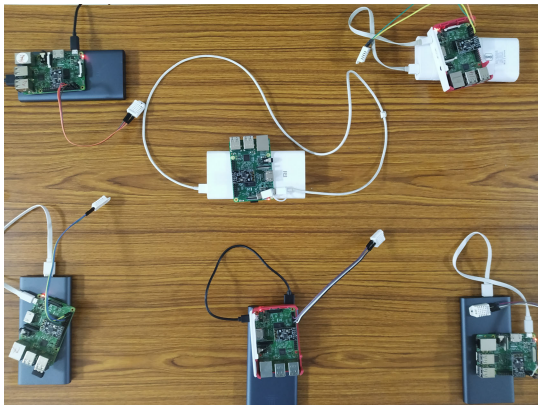


FIGURE 15. Test-bed setup with IEEE 802.15.4.

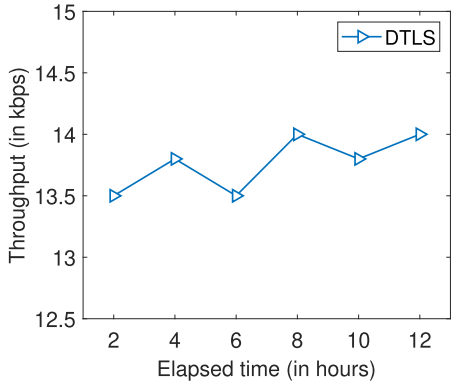


FIGURE 16. Throughput performance of the proposed scheme in a real testbed.

obtained through both sets of experiments corroborate with our findings on higher performance benefits. We observe a performance (in terms of throughput) difference of around 6% between results obtained through testbed implementation and simulation. This is due to interference from other operating networks during data transmission.

1) DISCUSSION ON REAL-WORLD DEPLOYMENT

As the traffic volume in real-world has drastically increased, automation of the current traffic system is necessary. The proposed system is easily scalable for a city by adding more devices equipped with RPis, camera modules, and low-power communicating modules. Once the camera module captures the image from a lane, the optimized Yolo computes the number of vehicles waiting in each lane, following which Algorithm 1 is executed as described in Subsection IV-C at the central junction device. The system can be integrated with the existing traffic light infrastructure (manually operated or timer-based) through IEEE 802.15.4 DSME-based communication technology. Algorithm 1 signals the appropriate traffic lane with green and the rest with red for F_{lane_i} time units. For computing the cumulative delay through important blocks (Algorithm 2), LoRa communication technology is used as it facilitates long-distance communication for transmitting the delay from one traffic junction to another. The delay for a lane is computed at each junction depending on the waiting time between two green signals. This time is transmitted over the LoRa signals by the central junction device that computes Algorithm 1 and is equipped with LoRa modules.

Additionally, due to the use of low-power technologies like IEEE 802.15.4 and LoRa, which are the de-facto IoT communication protocols, integration with any other smart city IoT application is possible. One of the primary design goals of the proposed system is to use technologies that support multiple applications. For example, the LoRa network server can host multiple smart city application servers.

VI. CONCLUSION & FUTURE WORK

In this paper, we presented a smart traffic signal system architecture that optimizes the cumulative delay incurred by vehicles at traffic junctions. Image approximation is applied using the RAKE algorithm for optimal and efficient object detection at different junctions. In this work, we have pruned the YOLO network during the training to cut out some weights or unimportant connections that do not affect and contribute to the final result. This facilitates low overhead during computation through IoT devices. Further, the proposed DTLS mechanism ensures that vehicles are prioritized based on the cumulative delay incurred. Also, a smart green corridor for emergency vehicles is considered in this mechanism. The use of DSME-GTSs and low-power link layer technologies (IEEE 802.15.4 and LoRa) results in minimal energy consumption by the proposed schemes while providing low latency and high throughput.

As a part of future work, we plan to enhance the employment of machine learning algorithms to predict traffic patterns based on historical data and real-time sensor data. This can be used to optimize the timing of traffic lights and reduce congestion. Additionally, connected vehicles can provide real-time data on traffic conditions, allowing traffic lights to adjust their timing in response to changing traffic patterns.

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